Credit Risk Monitor: how the system works

From raw files to the analyst's screen. Blue = data and model steps, purple = where the AI (Google Gemini) is used, orange = AI feature 2, green = the person in charge.

1. Raw data

Tool: CSV files in the repo

5 CSV files, never changed:

- customers: 10,000
- loans: 14,685 (with the answer: did it go bad?)
- loans at other banks: 22,979
- monthly payments: 176,220
- collection actions: 8,635

2. Clean and join

Tool: pandas, numpy (src/ingest.py)

- Fix labels, fill gaps with group medians and a 'was missing' flag, cap impossible values
- Squash 12 payment rows into 1 row per loan; other-bank rows into 1 per customer
- Join everything: **1 row per loan**, row count checked (14,685)
- Every fix is written to a data-quality log

3. Portfolio analytics

Tool: pandas (src/analytics.py)

- Key numbers, breakdowns by loan type, income, job, region, age
- Two-field cross-cuts (e.g. income x job)
- 5 insights with 'what to do'
- **One source of truth:** the charts and the AI use the same functions

4. Risk model

Tool: scikit-learn, XGBoost (src/model.py)

- Simple model (logistic regression) + main model (XGBoost)
- Trained on 75% of customers, tested on 25% it never saw
- Gives each loan a **chance of going bad** and its top reasons (SHAP)
- Not used as inputs: gender, marital status, IDs, dates, collections

5. Dashboard

Tool: Streamlit, Plotly (app/)

6 tabs: Portfolio, Account, Ask the Portfolio, Anomaly Review, Collections, Model & Monitoring

- Filters on the left, a floating 'Ask' button on every tab
- Each AI feature shows its trigger, false alarms, misses and human step
- Hosted on Streamlit Cloud, no login

Where it runs and how it is kept safe

- Build step: **python src/build.py** (clean, train, score, flag). App: **streamlit run app/app.py**
- Live app on Streamlit Community Cloud; code on GitHub; Python 3.12, packages pinned
- The AI key lives in Streamlit Secrets (on a laptop: a .env file that is never uploaded)
- No key, or the AI is down? The dashboard still works and ready-made reports still answer

AI feature 2: spotting unusual accounts (src/anomaly.py)

Tool: pandas, scikit-learn

Runs on every account each time data is refreshed. Flags an account when:

- **vs its own past:** paid on time for 9 months then missed, or days late jumped by 30+
- **vs similar customers** (same loan type and income group): in the top 1% for credit used or days late at other banks
- **rare mix:** Isolation Forest puts it in the rarest 2%. Each flag comes with a plain reason.

Result: 981 flagged.

Analyst decides

Confirm or Dismiss, with name and note.

Saved with the time. Nothing is deleted.

The system never acts on its own.

AI feature 1: Ask the Portfolio (src/llm_assistant.py), step by step

Analyst asks

Types a question, e.g. *'Which loan type goes bad most often?'*

AI call 1: pick a report

Gemini gets the question and the list of 7 approved reports. Replies with one report, or 'decline'.

Python works out the numbers

Our code checks the choice and runs it on the data (same functions as the charts).

AI call 2: write the answer

Gemini gets the question and **one small summary table**. Writes 1 to 3 sentences.

Python checks every number

Each number in the answer must match the table. If not, a safe answer is written by code instead.

Answer on screen

With a badge ('Checked against our data') and the source table to look at.

Model: gemini-3.5-flash (backup models if busy), temperature 0. The AI never calculates; it only chooses a report and words the answer. Forecasts or questions outside the data get a polite 'can't answer'.

What is sent to the AI service

Call 1: the analyst's question + the list of approved reports. **No data at all.**

Call 2: the question + one small summary table, e.g. 'share that went bad by loan type', or one account's risk and top reasons.

Never sent: raw files, the full dataset, customer names or rows of customer records, the API key.

People stay in control

Ask the Portfolio: the analyst checks the badge and the source table before using any number.

Anomaly Review: the analyst reads the reason and records Confirm or Dismiss with a note. Every decision is kept for audit.

Model: a person approves any retrain. Drift (PSI), accuracy and fairness are checked on the Model & Monitoring tab.